Heteroclinics and Permanent Rotations of the Suslov Body

Route-A source-local certificate HCS-C255

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Abstract

We give a complete positive-energy reduced atlas for the classical Suslov rigid body. Every generic energy ellipse has two explicit heteroclinics and two permanent rotations; the latter reconstruct as periodic $SO(3)$ attitudes. This source-local nonholonomic theorem has no arithmetic origin.

1 Constraint and reduced equations

Align the forbidden body axis with e_3 and rotate the allowed plane so that

$$I = \begin{pmatrix} I_1 & 0 & a \\ 0 & I_2 & b \\ a & b & I_3 \end{pmatrix}, \qquad I_1, I_2 > 0, \quad I_3 > \frac{a^2}{I_1} + \frac{b^2}{I_2}. \tag{1}$$

The constraint is $\omega_3 = 0$, and $I\dot{\omega} = (I\omega) \times \omega + \lambda e_3$. With $\ell = a\omega_1 + b\omega_2$, the first two components are

$$I_1\dot{\omega}_1 = -\ell\omega_2, \qquad I_2\dot{\omega}_2 = \ell\omega_1, \qquad H = \frac{1}{2}(I_1\omega_1^2 + I_2\omega_2^2). \tag{2}$$

Direct substitution gives $\dot{H} = 0$.

2 Both heteroclinic components

Set $u = \sqrt{I_1}\omega_1$, $v = \sqrt{I_2}\omega_2$ and $(u, v) = R(\cos\phi, \sin\phi)$, $R = \sqrt{2H}$. For

$$q^2 = \frac{a^2}{I_1} + \frac{b^2}{I_2}, \quad \cos\delta = \frac{a}{\sqrt{I_1}q}, \quad \sin\delta = \frac{b}{\sqrt{I_2}q}, \quad \kappa = \frac{Rq}{\sqrt{I_1 I_2}}, \tag{3}$$

equations (2) reduce to $\dot{\phi} = \kappa\cos(\phi - \delta)$.

Theorem 1. *If $q, H > 0$, the two components complementary to $\ell = 0$ are*

$$\sin(\phi - \delta) = \tanh\kappa(t - t_0), \qquad \cos(\phi - \delta) = \sigma \operatorname{sech}\kappa(t - t_0), \quad \sigma = \pm 1. \tag{4}$$

Both run from the reduced-unstable permanent rotation to its antipodal reduced-stable partner, with exponents $+\kappa$ and $-\kappa$.

The identities $\tanh^2 z + \operatorname{sech}^2 z = 1$ and differentiation of (4) prove the theorem. The scaled angular deflection is π.

3 Permanent rotations and boundaries

At $\ell = 0$, angular velocity is constant. The full attitude therefore is

$$Q(t) = Q_0 \exp(t\widehat{\omega}), \qquad T = \frac{2\pi}{|\omega|} \quad (\omega \neq 0). \tag{5}$$

Thus every generic energy ellipse already has periodic full-flow rotations; they must not be erased by the heteroclinic statement. If $a = b = 0$, every reduced state is permanent and every nonzero full state is periodic. If only one of a, b vanishes, (3)–(4) remain generic. At $H = 0$ the attitude is fixed.

4 Reduced Poisson structure and reversor

The two-dimensional reduced flow is generated by H through

$$\{\omega_1, \omega_2\} = -\frac{\ell}{I_1 I_2}. \tag{6}$$

Every bivector in dimension two satisfies Jacobi. On either open half-plane $\ell \neq 0$, division by $|\ell|$ leaves a divergence-free linear rotation, so $d\omega_1 d\omega_2/|\ell|$ is invariant. At a nonzero permanent rotation the original divergence is nonzero; consequently no positive C^1 invariant density can extend through that endpoint. The full involution $(Q, \omega) \mapsto (Q, -\omega)$ reverses time because the reduced quadratic vector field is even.

face	reduced behavior	full reconstruction
$q > 0$, $\ell \neq 0$	two heteroclinic arcs	nonrecurrent state
$q > 0$, $\ell = 0$	two permanent endpoints	periodic rotations
$q = 0$, $H > 0$	entire ellipse permanent	periodic rotations
$H = 0$	origin	fixed attitude

5 Executable boundary

The exact certificate stores twelve generic and four principal-axis inertia rows. Rational squares record the radical scales. The checker reconstructs Schur positivity, endpoints, exponents and periods; SymPy verifies (2)–(4), and replay plus rehashed semantic mutation closes integrity.

6 Route-A stop

The arithmetic origin is `none`. The tuple is `(A0_FAIL,A1_WEAK,A2_FAIL,A3_FAIL,A4_FORMAL_HINT)` and the verdict is `ROUTE_A_REJECTED`, with `route_b_invocation_allowed: false`, under `NO_BAD_EULER_OR_ROOT_NUMBER`. No target determinant or Hilbert–Pólya operator is claimed.

References

[1] L. C. García-Naranjo, J. C. Marrero, D. Martín de Diego and P. E. Petit Valdés, *J. Nonlinear Sci.* 34 (2024), article 110.

[2] Y. N. Fedorov, A. J. Maciejewski and M. Przybylska, *Nonlinearity* 22 (2009), 2231–2259.

[3] Y. N. Fedorov and V. V. Kozlov, *AMS Transl.* 168 (1995), 141–171.